Homework 4

February 19 - 26

Reading

1. CS229 Probability Reference: Sections 1–4.1. Some of this may be review, but I'd still like you to go through and make sure you understand it. Their treatment of continuous random variables is a bit sparse/terse, so if you end up finding yourself lost at all, take a look at R+G 2.4, which is a bit gentler of an introduction.

2. Rogers and Girolami: Sections 2.2–2.5, excluding 2.5.2 and 2.5.4. I want you to *very lightly* skim these sections after reading the CS229 notes, making sure you understand it all. You can also take a look at these sections if you get stuck on the CS229 notes for a gentler introduction to some topics.

2. Rogers and Girolami: Sections 2.1, 2.6–2.8.

3. CS229 Notes: If you have the time, go back and finish Section 8.1 (you can skip 8.1.1 if you want, but it is pretty interesting).

Problems

Problem 1 (Conditional Probability). Let X, Y, Z be random variables.

(a) Prove the identity $p(x \mid y, z) = \dfrac{p(x, y \mid z)}{p(y \mid z)}$.

(b) We say two random variables A and B are *conditionally independent* on the random variable C if $p(a, b \mid c) = p(a \mid c) \cdot p(b \mid c)$. If X and Y are conditionally independent on Z, show that

$$p(x \mid y, z) = p(x \mid z).$$

Hint: Use part (a).

Problem 2. R+G gives the identities on the top of page 69 without much justification. In this problem we'll prove them ourselves. Put $n = 1960$ for brevity. As the book mentions, we need to operate under the following independence assumptions:

1. All targets t_i are conditionally independent on the joint input distribution $(\mathbf{x_n}, \mathbf{X})$
2. The targets $\mathbf{t}$ and the input $\mathbf{x_n}$ are conditionally independent on the remaining input $\mathbf{X}$.
3. The target t_n and the inputs $\mathbf{X}$ are conditionally independent on the single input $\mathbf{x_n}$

Under these assumptions, prove the following.

(a) $p(t_n \mid \mathbf{x}_n, \mathbf{X}, \mathbf{t}) = \dfrac{p(t_n, \mathbf{t} \mid \mathbf{x}_n, \mathbf{X})}{p(\mathbf{t} \mid \mathbf{X})}$.

(b) Use part (a) to show that $p(t_n \mid \mathbf{x}_n, \mathbf{X}, \mathbf{t}) = p(t_n \mid \mathbf{x}_n)$.

Hint: Use Problem 1.

Problem 3 (Hessian Matrix). For a vector-valued function $f : \mathbb{R}^n \to \mathbb{R}^m$, we often denote f by its *component functions* $f_i : \mathbb{R}^n \to \mathbb{R}$, where $f(x) = [f_1(x) \ \cdots \ f_m(x)]^T$. We define f's *Jacobian* at a point $a \in \mathbb{R}^n$ to be the matrix containing the partial derivatives of its component functions at a:

$$J_f(a) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(a) & \cdots & \frac{\partial f_1}{\partial x_n}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(a) & \cdots & \frac{\partial f_m}{\partial x_n}(a) \end{bmatrix} = \begin{bmatrix} \nabla f_1(a)^T \\ \vdots \\ \nabla f_m(a)^T \end{bmatrix}$$

For a real-valued function $f : \mathbb{R}^n \to \mathbb{R}$, as in Comment 2.6 of R+G, we define its *Hessian* at the point $a \in \mathbb{R}^n$ to be the $n \times n$ matrix

$$H_f(a) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(a) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(a) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(a) \end{bmatrix}.$$

(a) Convince yourself that $H_f(a) = J_{\nabla f}(a)$. That is, that the Hessian is the Jacobian of the gradient.

(b) Fix $A \in M_{m \times n}$ and $b \in \mathbb{R}^m$ and define $f : \mathbb{R}^n \to \mathbb{R}^m$ by $f(x) = Ax + b$. Prove that $J_f(a) = A$ for every $a \in \mathbb{R}^n$.

(c) Here, we assume the same setup as Problem 4 of Homework 1. The log likelihood is then defined as

$$\log L = -N \log \sigma - \frac{N}{2} \log 2\pi - \frac{1}{2\sigma^2} \sum_{n=1}^{N} (y^{(n)} - \theta^T x^{(n)})^2.$$

Taking $\log L$ to be a function of θ, show that $\nabla_\theta \log L = \frac{1}{\sigma^2}(X^T y - X^T X\theta)$. Use this identity and parts (a) and (b) to verify Equation 2.38:

$$H_{\log L}(\theta) = -\frac{1}{\sigma^2} X^T X.$$

Coding

Go make an account on TensorTonic. This looks like a super cool resource to practice coding up some ML concepts. They also have some modules under "ML Math" that you might be interested to look at to supplement your own learning. Take some time to explore! Below are some problems you should take a look at. You don't need to do all of them—feel free to pick and choose. Take this as an opportunity to start learning some basic numpy.

1. Linear Algebra. Implement Dot Product (Easy), Implement Euclidean Distance (Easy), Matrix Transpose (Easy), Matrix Inverse (Medium).

2. Basic ML. Mean Squared Error (MSE) (Easy), Implement Gradient Descent for a 1D Quadratic (Easy), Linear Regression Closed Form (Medium).

Fun Stuff

1. Watch: "The most complex model we actually understand" - Welch Labs